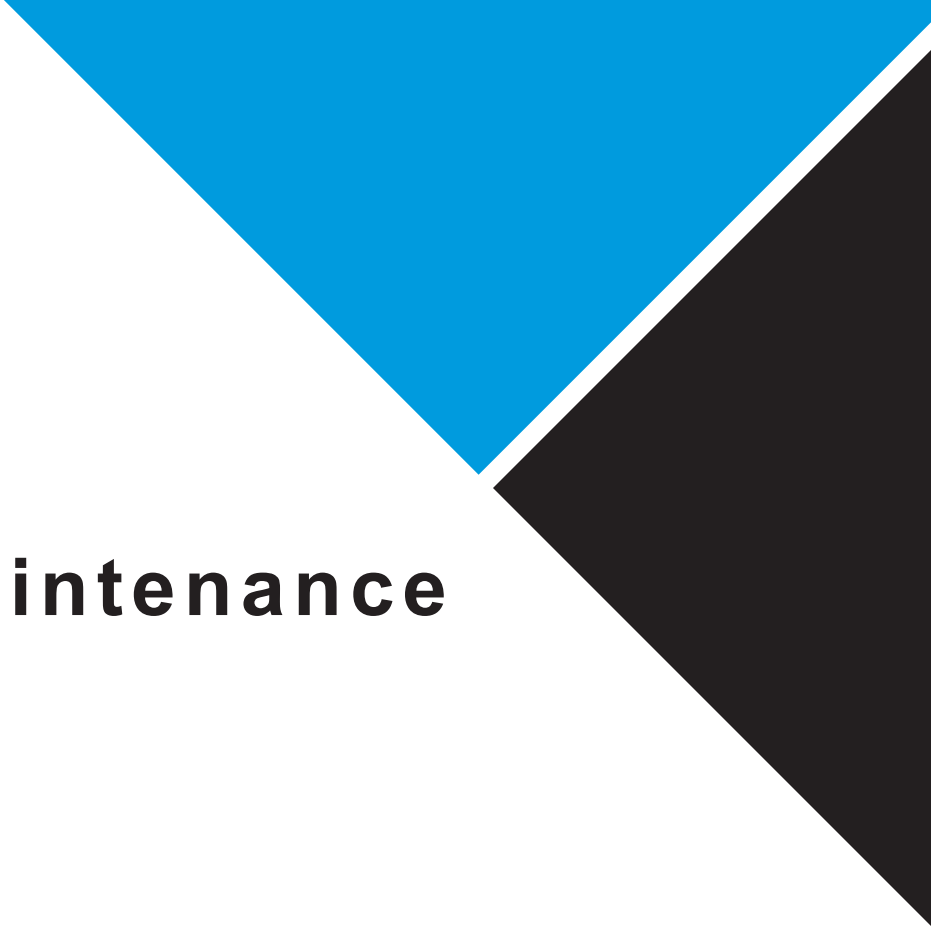




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Maintenance

Machine Maintenance

The maintenance schedule provides the maintenance task number along with a description of the task and the interval in hours. Each task should be performed at every interval indicated, not just the first instance. For the convenience of the end-user, the following table is provided to cross-reference the hourly interval to alternate equivalent time units. Some of the alternate equivalency intervals are approximate.

NOTICE

The following table presumes an eight-hour shift. If the end-user deploys the equipment on a schedule that features either more or less shifts per day, then the alternate equivalent intervals must be adjusted accordingly.

NOTICE

If it is necessary to have repair work performed under warranty, it will help to have proof that regular maintenance work was performed and that the damage was not caused by insufficient or improper maintenance.

NOTICE

The machine parts should be replaced by a trained technician who is familiar with the machine and good practices.

NOTICE

For more in-depth treatment of maintenance matters, BluePrint Automation offers a comprehensive Preventative Maintenance Program which is available from our service department.

⚠ WARNING

Before any maintenance or repair work being performed on the machine, standard Lockout/Tag-out procedures should be followed according to internal policy.

⚠ WARNING

Only qualified service technicians may perform maintenance and repair work on the machine.



COLLATOR 300
maintenance video



HOURLY INTERVAL	ALTERNATE EQUIVALENT
8	Shift
16	Daily
96	Weekly
300	Triweekly
400	Monthly
800	Bimonthly
1250	Trimonthly
1600	Quarterly
2500	Semi-annually
5000	Annually
10 000	Biennially
20 000	Quadrennially

MAINTENANCE TASKS	ACTION	Hours
CLEAN THE MACHINE THOROUGHLY	CLEAN	8
CHECK FOR LOOSE PARTS	TIGHTEN	8
CHECK FOR MISSING, DAMAGED, OR WORN PARTS	REPLACE	8
CHECK FOR ANOMALOUS NOISES	ADJUST	8
CHECK THE TRANSPORT BELTS TRACKING AND WEAR	ADJUST	8
CHECK THE DRIVE BELTS FOR WEAR	ADJUST	8
CHECK TRANSPORT BELTS TENSION	ADJUST	96
CHECK DRIVE BELTS TENSION	ADJUST	96
CHECK BEARINGS AND SEALS FOR MISALIGNMENT	ADJUST	96
CLEAN ALL SENSOR HEADS AND REFLECTORS	CLEAN	96
CHECK GEAR-HEAD VENT PLUGS FOR OBSTRUCTIONS	CLEAN	96
CLEAN CABINET COOLING FAN FILTER ELEMENTS	CLEAN/REPLACE	400
CLEAN AND LUBRICATE LEAD-SCREWS	CLEAN/LUBRICATE	400
CHECK GEAR-HEADS FOR OIL LEAKS	REPAIR/REPLACE	400
CLEAN ELECTRIC MOTOR COOLING FAN VENTS	CLEAN	1250
CHECK BEARINGS AND SHAFTS FOR EXCESS WEAR	ALIGN/REPLACE	1600
CHECK LUBRICATION IN GEAR-HEADS	LUBRICATE	2500
REPLACE THE COUNT BELT	REPLACE	2500
CHECK MOTOR COOLING FANS FUNCTIONALITY	CHECK	2500
REPLACE THE DRIVE BELTS	REPLACE	5000
REPLACE THE TRANSPORT BELTS	REPLACE	5000
REPLACE THE FLANGE MOUNTED BEARINGS	REPLACE	5000
REPLACE THE RADIAL BALL BEARINGS	REPLACE	5000
REPLACE LUBRICANT IN THE GEAR-HEADS	REPLACE	10000
REPLACE THE GEAR-MOTORS SHAFT SEAL RINGS	REPLACE	10000
RE-LUBRICATE GEAR-MOTOR BEARINGS	LUBRICATE	20000

Daily Inspection

For safe and optimised machine performance, it is good practice to regularly inspect the machine for cleanliness, missing, worn, or damaged parts as well as the efficacy of all safety features.

Routine Cleaning

The following cleaning supplies are suggested:

- Clean damp cloth.
- Soft bristled brush.
- General water-based butyl cleaner/degreaser (such as ProChem® Indusol or Würth Workplace Cleaner Absobon®).

Remove all product and packaging materials from the machine and clean the interior surfaces, taking care to remove any accumulated product or adhæsive residues.

Clean the operator's HMI screen (with HMI off).

Routine Inspection

At a minimum, the following items should be checked for wear, damage, or corrosion etcætera:

- Check pneumatic valves and cylinders for efficacy.
- Check all moving parts for wear or damage.
- Check components that come in contact with the product or packaging material.
- Check door hardware integrity and safety switch efficacy and all other safety components for good functioning.

NOTICE

Any defects or needed repairs observed by the operator which are outside the operator's purview are to be reported immediately to the maintenance department.

! WARNING

Before any maintenance or repair work being performed on the machine, standard Lockout/Tag-out procedures should be followed according to internal policy.

Recommended Products for Maintenance		
Application	Category	Product name
NORD gear-heads	Oil	ExxonMobil 600 XP220®
NORD gear-heads	Oil	ExxonMobil SHC® 636
Flat-head screws in blind holes	Adhæsive	Henkel LOCTITE® 242
Threaded connexions (non-stainless)	Adhæsive	Henkel LOCTITE® 243
Bearing and taper-lock set-screws	Adhæsive	Henkel LOCTITE® 248
Cylindrical fitting parts	Adhæsive	Henkel LOCTITE® 609
Gear-head shaft bores	Grease	Henkel LOCTITE® LB 8014
Stainless steel screw threads	Oil	Henkel LOCTITE® LB 8060
Housed bearings	Grease	Henkel LOCTITE® LB 8104
NORD gear-head roller bearings	Grease	Klüber Petamo® GHY 133 N
NORD gear-heads	Oil	Klübersynth GH 6-220®
General cleaning	Cleaner	ProChem® Indusol 2214
General cleaning	Cleaner	Würth Workplace Cleaner Absobon®

Standard Cleaning Method

The standard version of the machine is not manufactured for wash-down applications, and must not be cleaned with water.

Required Equipment

The following items are recommended for cleaning machines with standard environmental exclusion rating.

- Vacuum cleaner.
- Non-abrasive lint-free cloth.
- PPE per internal policy.

Cleaning Instructions

1. Use suction to remove all loose particles.



2. Use a dry or moistened (light detergent may be used) lint-free cloth to wipe the dirt and dust from the machine surfaces.



DANGER

Turn off and lockout all electrical power and fluid power sources per internal policy before entering the workspace.
Also read the safety section of this manual.

CAUTION

Food-grade Lubricated Models: Make sure that no liquid flows into the machine or stagnates in any gap or surface after cleaning.

NOTICE

Following the wrong cleaning procedures can, and likely will result in damage to the machine.

Polycarbonate cleaning

These cleaning recommendations apply to all polycarbonate sheet products. Periodic cleaning using correct procedures can help to prolong service life.

NOTICE

Note: Never use products made for cleaning glass (windows) or made with alcohol.

- Do not clean polycarbonate in direct sunlight or at high temperatures as this can lead to staining.
- Do not use chemical bonding agents on fasteners in contact with polycarbonate panels, these will cause the polycarbonate to crack.

Cleaning Procedure for Small Areas

1. Gently wash the polycarbonate sheet with a solution of mild soap and lukewarm water, using a soft, lint-free cloth or sponge to loosen any dirt or grime.
2. Fresh paint splashes, grease and smeared glazing compounds can be removed easily before drying by rubbing lightly with a soft cloth using petroleum ether (BP65), hexane or heptane. Afterwards, wash the sheet using mild soap and lukewarm water.
3. Scratches and minor abrasions can be minimised by using a mild automobile polish. We suggest that a test be made on a small area of the sheet with the polish selected and that the polish manufacturer's instructions be followed, before using the polish on the entire sheet.
4. Finally, thoroughly rinse with clean water to remove any cleaner residue and dry the surface with a soft cloth to prevent water spots.

Other Important Instructions for All Polycarbonates:

- Never use abrasive or highly alkaline cleaner.
- Never use aromatic or halogenated solvents like toluene, benzene, gasoline, acetone or carbon tetrachloride on polycarbonate materials.
- The use of incompatible cleaning materials with polycarbonate (Lexan) sheet can cause structural and/or surface damage.
- Contact with harsh solvents such as methylethylketone (MEK) or hydrochloric acid can result in surface degradation and the possible crazing of the polycarbonate.
- Never scrub with brushes, steel wool or other abrasive materials.
- Never use squeegees, razor blades or other sharp instruments to remove deposits or spots.

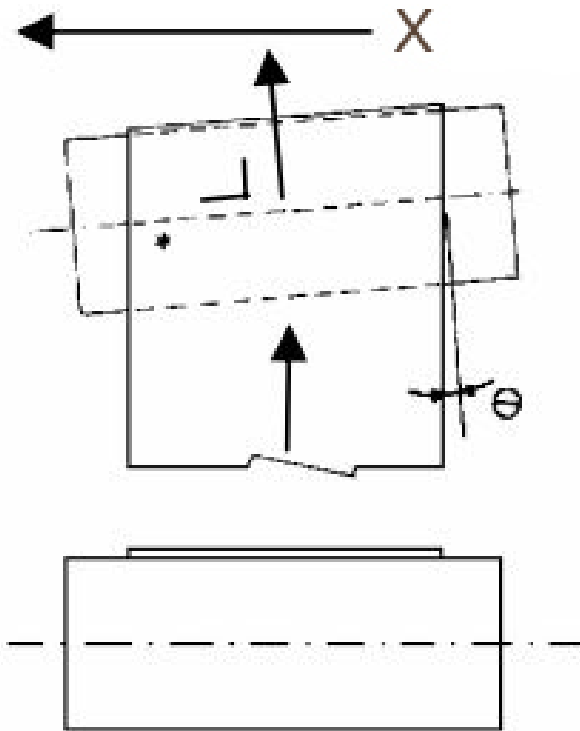
Conveyor Belt Maintenance

A belt running against machine parts because of mis-tracking results in fraying edges, shorter belt life, and possibly other damage to drive components. The particles from deteriorating belts can eventually contaminate the product. This can be prevented by proper belt tensioning and tracking.

Check conveyor belts for wear, fraying, tracking and tension. Check lacing or other joining technologies for integrity. Replace any unserviceable belts.

Conveyor Belt Tracking

When a pulley is not placed square to the belt, it will result in the situation as shown below. The belt will move to the left (X).



The problem is corrected by restoring the perpendicularity of the pulley to the belt.

NOTICE

Transport belts with guides on the drive contact surface (underside) do not require tracking. Attempting to adjust tracking on guided transport belts can result in premature belt failure.

Conveyor Belt Tension

Belt tension must be great enough to prevent slippage between the drive pulley and belt.

Slippage will cause excessive wear to both the drive pulley lagging and the belt.

Proper minimum belt tension is the tension required so that a given belt conveyor or belt elevator system will operate properly in its environment.

Operate the belt conveyor system under the most adverse conditions it will encounter (i.e., under load), and observe closely to see if there is any slippage between the drive pulley and the belt. If there is, more tension is required.

Do not over tension, minimum belt tension is just enough, not too much.

Conveyor Belt Installation

1. Do not pry or force the belt onto the pulleys.
2. Reduce the pulley's centre distance so that the belt can be easily installed.
3. Place the belt on the pulleys; do not pry or force the belt.
4. Increase the belt tension until the belt feels snug or taut. Avoid over tensioning the belt.
5. Start the drive under the most severe load conditions. This may be either the motor starting torque or during the work cycle. A belt that is too loose will jump teeth at this time.
6. Tighten the belt gradually until a satisfactory operating condition is achieved.

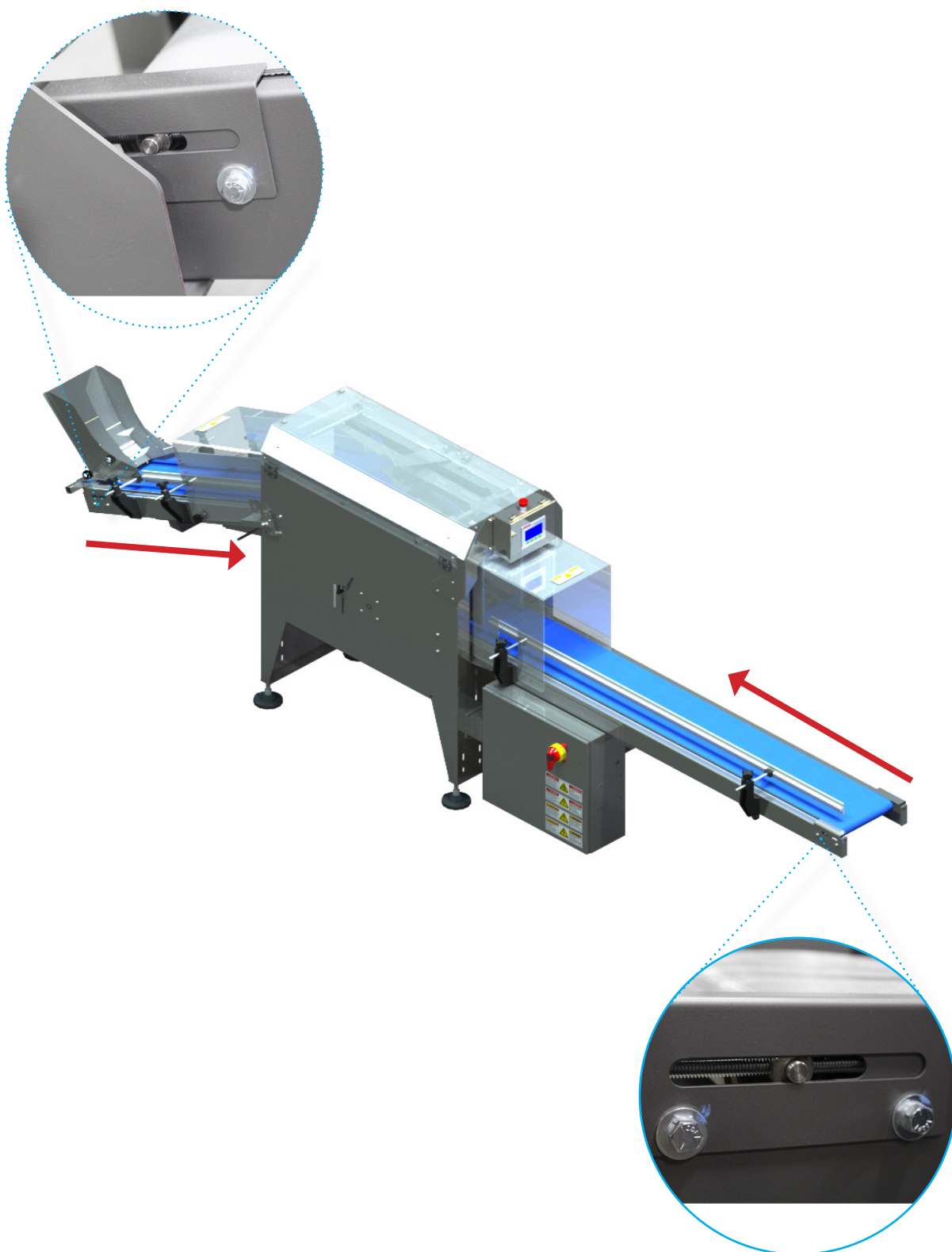
In-feed and Exit Belts Maintenance

To adjust or replace the in-feed or exit conveyor belts you must act on the various belt tensioners.



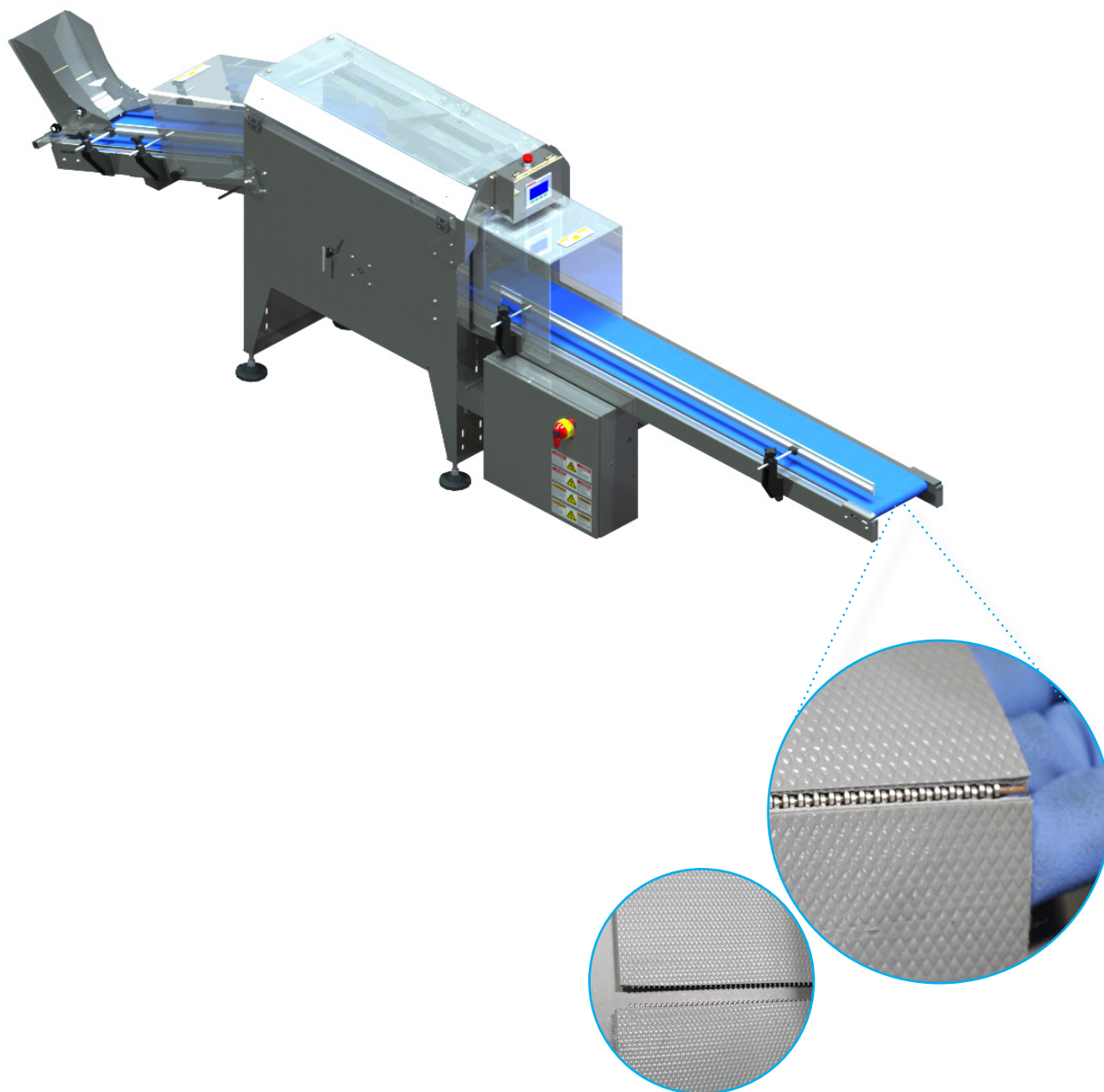
1

Loosen the two tension screws at the end of the shaft; there are two sets, one set on the in-feed belt located under the in-feed chute and the other set is located at the end of the exit belt.



2

Adjusting the screws will move the shaft back in toward base of machine as the red arrows indicate above, allowing the conveyor belt to loosen and have slack.



3

Locate where the belt binds together and remove the joining pin; allowing the ends to separate. Reassemble and apply tension via the tensioners.

Count Belt Maintenance

To adjust or replace the count conveyor belt, follow the steps indicated below.



1

With the belt installed, use a 5/32" hexagon key, loosen the four 1/4" hexagon socket countersunk screws as shown above.



2

Use a lever to adjust the weldment; applying pressure with the lever will slide the weldment back and forth. Make sure to adjust both sides of the weldment equally, maintaining parallelism while at the same time, making sure to keep the in-feed belt level with the package guide.



3

Make sure to go by the one finger rule; you should be able to put one finger in between the conveyor belt and the weldment. Once this is confirmed and in-feed belt is adjusted to the desired tension, tighten the four 1/4" hexagon countersunk head screws with a 5/32" hexagon key.

Synchronous Belt Drives Maintenance

Check for damage, fraying, or wear, replace any unserviceable belts. Check tension and alignment per paragraphs below.

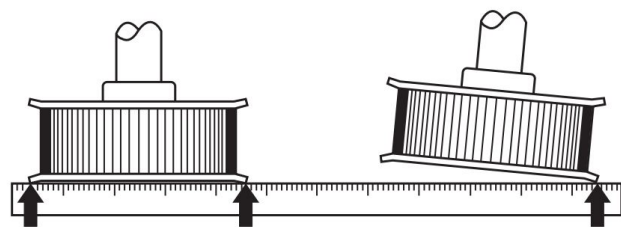
Belt Tension

Synchronous belt drives generally require little if any re-tensioning when used in accordance with proper design procedures. A small amount of belt tension decay can be expected within the first several hours of operation (running-in period). After this time, the belt tension should remain relatively stable.

Pulley alignment

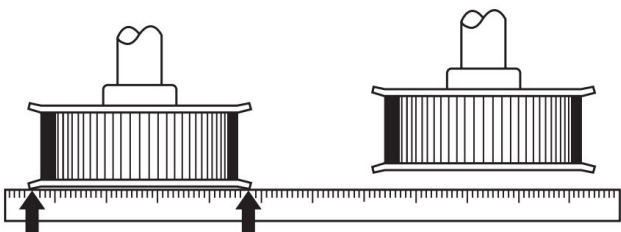
Drive misalignment is one of the most common sources of drive performance problems. Misaligned drives can exhibit symptoms such as high belt tracking forces, uneven belt tooth wear, high noise levels, and tensile cord failure. The two primary types of drive misalignment are angular and parallel. Discussion about each of these types are as follows:

Angular misalignment results when the drive shafts are not parallel. As a result, the belt tensile cords are not loaded evenly, resulting in uneven tooth/land pressure and wear. The edge cords on the high tension side are often overloaded which may cause an edge cord failure that propagates across the entire belt width.



Angular misalignment often results in high belt-tracking forces as well which cause accelerated belt edge wear, sometimes leading to flange failure or belts tracking off of the pulleys.

Parallel misalignment results from pulleys being



mounted out of line from each other. Parallel misalignment is generally more of a concern with V-type belts than with synchronous belts because V-type belts run in grooves and are unable to free float on the pulleys.

Synchronous belts will generally free float on the pulleys and essentially self-align themselves as they run. This self-aligning can occur as long as the pulleys have sufficient groove face width beyond the width of the belts. If not, the belts can become trapped between opposite pulley flanges causing serious performance problems. Parallel misalignment is not generally a significant concern with synchronous drives as long as the belts do not become trapped or pinched between opposite flanges.

Allowable Misalignment: In order to maximize performance and reliability, synchronous drives should be aligned closely. This is not, however, always a simple task in a production environment. The maximum allowable misalignment, angular and parallel combined, is $1/4^\circ$.

Angular misalignment is not always easy to measure or quantify. It is sometimes helpful to use the observed tracking characteristics of a belt, to make a judgment as to the system's relative alignment. Neutral tracking "S" and "Z" synchronous belts generally tend to track "down hill" or to a state of lower tension or shorter centre distance when angularly misaligned. This may not always hold true since neutral tracking belts naturally tend to ride lightly against either one flange or the other due to numerous factors discussed in the section on belt tracking. This tendency will generally hold true with belts that track hard against a flange. In those cases, the shafts will require adjustment to correct the problem.

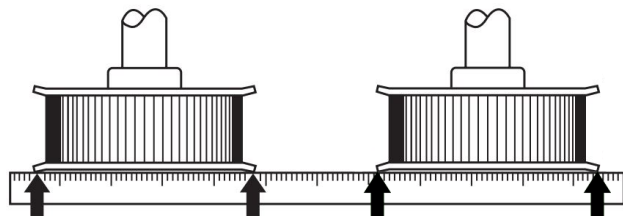
Parallel misalignment is not often found to be a problem in synchronous belt drives. If clearance is always observable between the belt and all flanges on one side, then parallel misalignment should not be a concern.

Belt Replacement

During the belt installation process, it is very important that the belt be fully seated in the pulley grooves before applying final tension. Serpentine drives with multiple pulleys and drives with large pulleys are particularly vulnerable to belt tensioning problems resulting from the belt teeth being only partially engaged in the pulleys during installation. In order to prevent these problems, the belt installation tension should be evenly distributed to all belt spans by rotating the system by hand. After confirming that belt teeth are properly engaged in the pulley grooves, belt tension should be rechecked and verified. Failure to do this may result in an under-tensioned condition with the potential for belt ratcheting.

Belt Removal

1. Loosen and remove all screws from the bushing.
2. Put the screws in the threaded holes in the outer piece of the assembly.
3. Tighten the screws equally to force the mating part loose from bushing.
4. After replacing the pulley, the shafts the shafts must be parallel and the pulleys must be coplanar to ensure proper wear and extending belt life.
5. Place a straight edge against the pulleys.
6. Make certain the straight edge touches all four points of the pulleys. The straight edge should be at the widest possible distance on the pulleys.

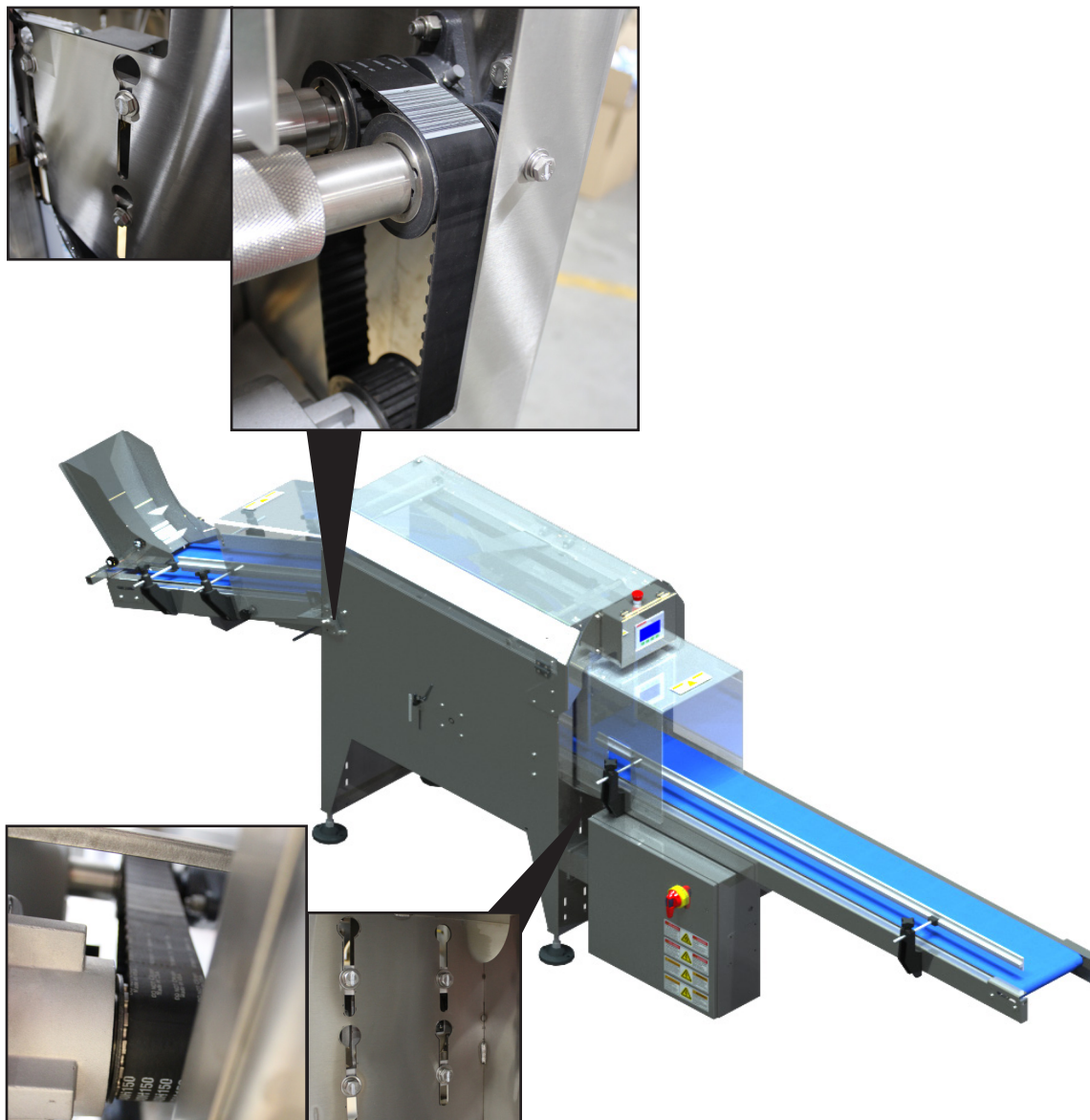


Belt Installation

1. Do not pry or force the belt on the pulleys.
2. Reduce the pulleys' centre distance so the belt can be easily installed.
3. Place the belt on the pulleys. Do not pry or force the belt.
4. Increase the belt tension until the belt feels snug or taut. Avoid over tensioning the belt.
5. Start the drive under the most severe load conditions. This may be either the motor starting torque or during the work cycle. A belt that is too loose will jump teeth at this time.
6. Tighten the belt gradually until a satisfactory operating condition is achieved.

Synchronous Drive Belts Maintenance

To adjust or replace the synchronous drive belts, follow the steps indicated below.



1

Loosen the four hexagon head cap screws that mount the motor to the machine housing just enough to allow the motor to slide vertically within the slots. Now the motor can be slid up in the slots thereby loosening the belt. Tighten the screws to keep the motor from dropping while the belt is exchanged (if necessary). To adjust the tension, loosen the screws allowing the weight of the motor to apply downward force thus tensioning the belt. Apply some additional downward force to the motor until optimum tension is obtained - approximately 3-6 mm of deflection. Re-tighten the screws.

Motor Fan Cleaning

Cleaning the cooling fan of fan cooled motors is an important maintenance task to ensure efficient cooling and prevent overheating. Here are the instructions for cleaning the cooling fan:

1. Safety precautions:

- Before starting any maintenance task, make sure the motor is completely powered off and disconnected from the power source to avoid electrical hazards.
- Allow sufficient time for the motor to cool down before cleaning to prevent burns.



Before any maintenance or repair work being performed on the motors, standard Lockout/Tag-out procedures should be followed according to internal policy.



Allow sufficient time for the motor to cool down before cleaning to prevent burns.

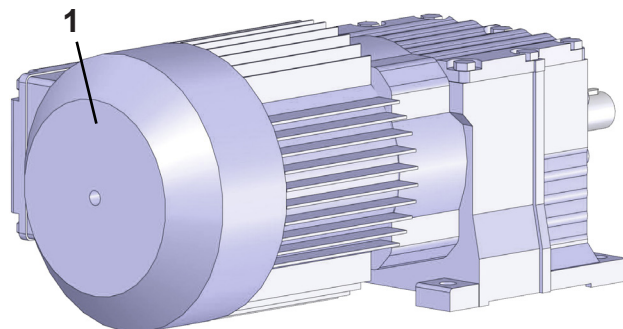


2. Gather the necessary tools:

- Soft-bristle brush.
- Compressed air gun.
- Vacuum cleaner or shop vacuum.
- Clean, lint-free cloth.
- Safety goggles and gloves (recommended).

3. Locate the cooling fan:

- The cooling fan is typically located at the non-drive end of the motor. Look for a protective cover or grille (1) that shields the fan blades.



4. Remove the fan cover:

- If the motor has a removable cover, carefully unscrew or unclamp it using appropriate tools.
- Some motors may have a hinged cover that can be opened; in that case, release the latch or hinge mechanism to access the fan.

5. Inspect the fan:

- Before cleaning, visually inspect the fan blades for any signs of damage or debris accumulation.
- If you notice any significant damage or distortion, replace damaged components.

NOTICE

Any defects or needed repairs observed by the technician which are outside the technician's purview are to be reported immediately to the technician's supervisor.

6. Remove loose debris:

- Using a soft-bristle brush, gently remove any loose dirt, dust, or debris from the fan blades, fan housing, and surrounding areas.
- Compressed air can be used on the fan guard, do not blow shop air toward the motor as this may force contaminants and debris into the motor housing.



- Be cautious not to bend or damage the blades during this process.
7. Vacuum out remaining debris:
 - Carefully use a vacuum cleaner or shop vacuum with a brush attachment to remove any remaining dirt or debris from the fan blades and housing.
 8. Wipe down the fan:
 - Moisten a clean, lint-free cloth with a mild detergent or a mixture of water and household cleaner.
 - Gently wipe down the fan blades and housing to remove any stubborn dirt or residue.
 - Avoid excessive moisture that may penetrate the motor; the cloth should be slightly damp, not dripping wet.
 9. Dry the fan:
 - Allow the fan blades and housing to air dry completely.
 - Make sure there is no residual moisture before reassembling the motor.
 - Reassemble the motor:
 - Once the fan blades are dry, carefully place the protective cover or grille back in its original position.
 - Secure it using the appropriate fasteners or clips.
 10. Final checks:
 - Double-check that all fasteners are tight and secure.
 - Ensure there are no loose tools or debris left near the motor.
 - Reconnect the motor to the power source and confirm proper functioning.

Regularly cleaning the cooling fan of a VFD motor will help maintain optimal cooling performance.

Rolling Bearing Maintenance Lubrication

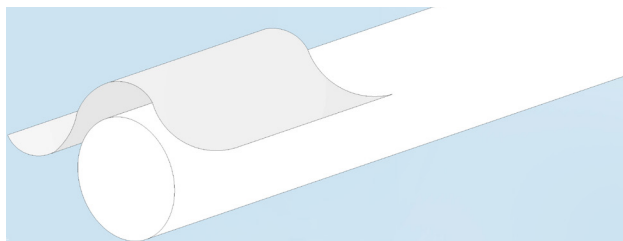
BluePrint Automation ball and roller bearing installations are considered lubricated for life and are therefore not furnished with grease fittings. Rather, we advise changing them out at their expected mean time to failure (MTTF).

Replacement

When replacing bearings, use appropriate tools (pullers, impact sleeves, impact rings, mandrels, lead-in bushings, presses, etc.). Do not pry bearings or apply uneven forces.



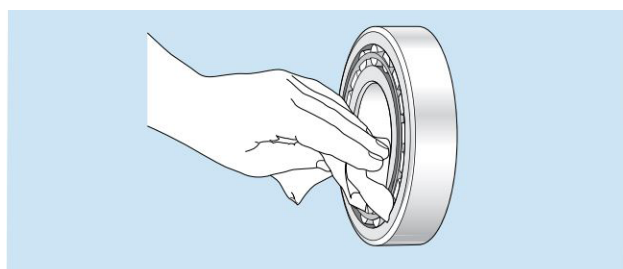
1. Make sure to burr the shafts and bores with an emery cloth or fine file.



2. Wipe the shaft with a clean cloth. New bearings do not need to be cleaned.



3. Wipe residual preservative from the bore of the bearing

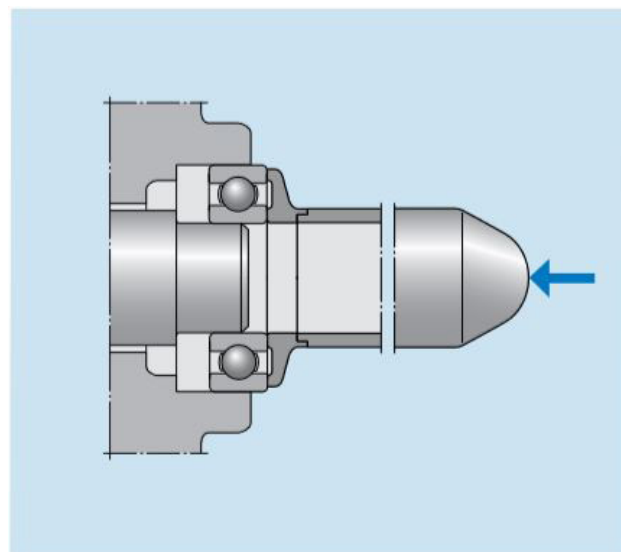


On mounting of non-separable bearings, the mounting forces must always be applied to the ring which will have the tight fit and therefore is the first to be mounted. Forces applied to the ring with the loose fit would be transmitted by the rolling elements, thus damaging raceways and rolling elements.

Mounting of separable bearings is easier, since the two rings can be mounted separately. In order to avoid score marks during assembly, slightly rotate the parts.

The use of a mounting disc is advisable when mounting simultaneously to shaft and housing assembly so that mounting forces are applied uniformly to inner and outer bearing races.

If a non-separable bearing is to be pressed onto the shaft and into the housing bore at the same time, the mounting force has to be applied equally to both rings at the same time and the abutment surfaces of the mounting tool must lie in the same plane. In this case a bearing fitting tool should be used, where an impact ring abuts the side faces of the inner and outer rings and the sleeve enables the mounting forces to be applied centrally.

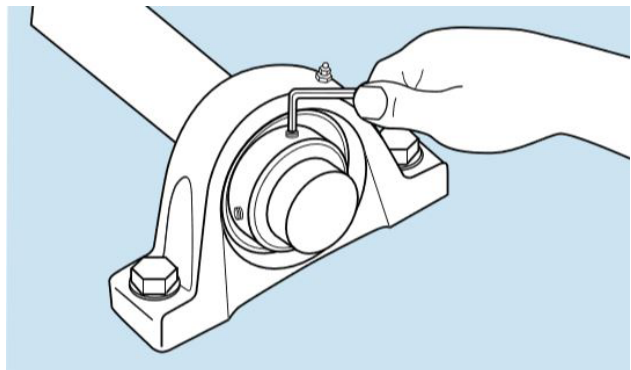


Housed Bearing Maintenance Lubrication

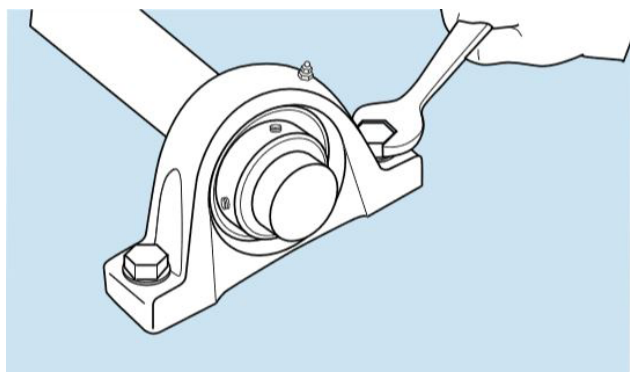
Generally speaking, pillow block and flange mounted bearing units are designed to operate without re-lubrication under normal speed and operating conditions. All ball bearing units are sealed at both sides with rubbing contact seals and are filled with a special long life grease of NLGI consistency 2. The grease has good corrosion inhibiting properties and is suitable for operating temperatures between -4°F and $+248^{\circ}\text{F}$.

Some bearing units may be equipped with a grease fitting that allows the bearing to be re-lubricated in service. When re-lubricating, care must be taken to use greases that are compatible with the original grease. We suggest a medium temperature, lithium calcium base, NLGI 2 grease having a base oil with a viscosity of 900 SUS (200 mm²/s) at $+100^{\circ}\text{F}$ ($+40^{\circ}\text{C}$). When a unit is being re-lubricated, avoid excessive pressure, which may cause damage to the bearing seals.

Dismounting instructions

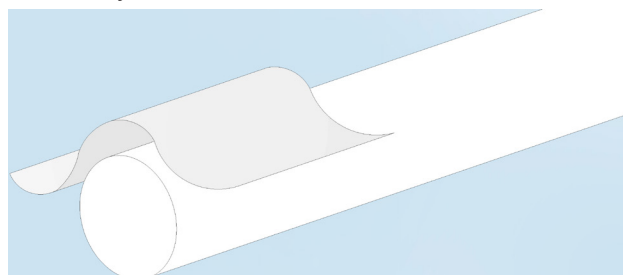


1. Loosen set-screws.
2. Unbolt the housing from its support. The complete bearing unit can then be removed from the shaft. It will be necessary to relieve the bearing load when removing the unit.



Mounting instructions

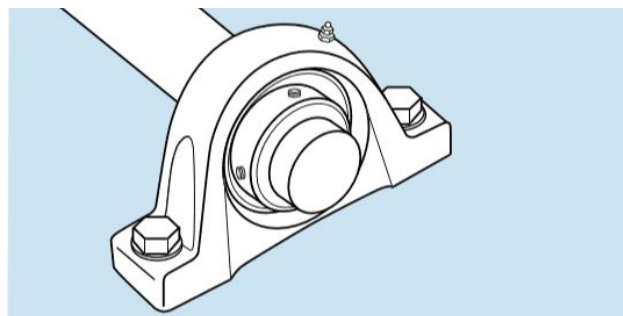
1. Remove any burrs or rust on the shaft with an emery cloth or a fine file.



2. Wipe the shaft with a clean cloth.



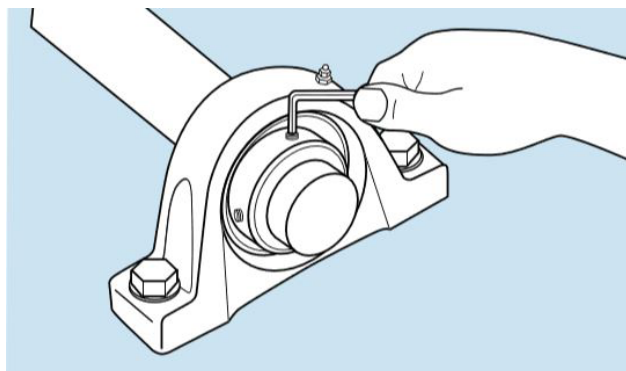
3. Check the shaft diameter
Recommended shaft tolerances:
Shaft diameter tolerance Up to 1 15/16" nominal to (49.2 mm) -0.0005" (-0.013 mm)
2" to 4" Nominal to (50.8 to 101.6 mm) -0.0010" (-0.025 mm)
4. Slide the bearing and housing onto the shaft and position. For eccentric lock-type units, leave the collar loose on the shaft.



5. Clean the base of pillow block and the support surface on which it rests. Be sure the supporting surface is flat. If the pillow block elevation must be adjusted by shims, the shims *must* extend the full length and width of the support surface. Bolt the pillow block securely to the support. With flanged housings, clean the flange and support surface. Be sure the support surface is flat. Bolt the flanged housing securely to the support.

1. Tighten each set-screw alternately with proper hexagon head socket wrench until they stop turning and the hexagon head socket wrench starts to spring. The spring of the hexagon head socket wrench can be easily seen and felt when the extension is used. When both set-screws are tightened on the shaft, the bearing is firmly seated. This completes the procedure for mounting the set-screw lock collars..See torque values.

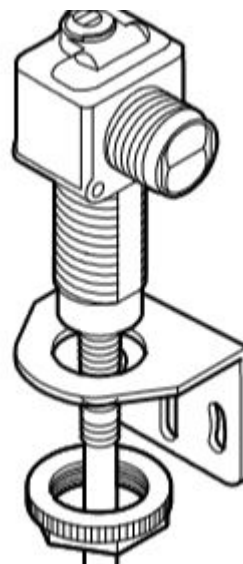
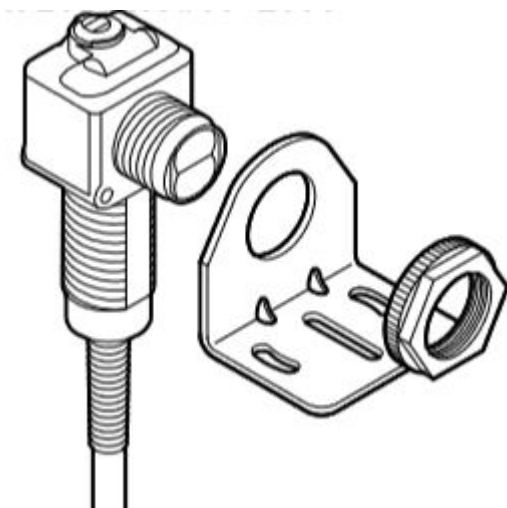
Torque Table	
Set-screw size	Torque in·lb (Nm)
#10-32 (M5)	36 (4.0)
1/4-28 (M6)	87 (9.8)
5/16-24 (M8)	165 (18.6)
3/8-24 (M10)	290 (32.8)
7/16-20 (M12)	430 (48.6)
1/2-20 (M14)	620 (70.1)



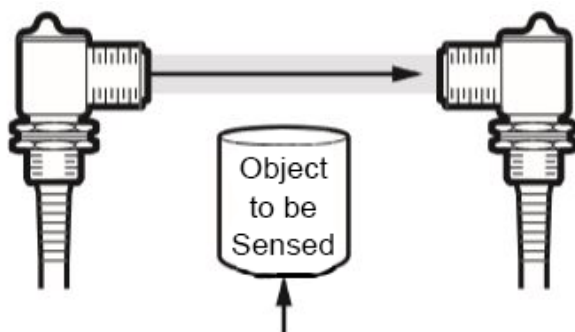
Photoelectric Sensors Maintenance

Various photoelectric sensors are used for product or position detection.

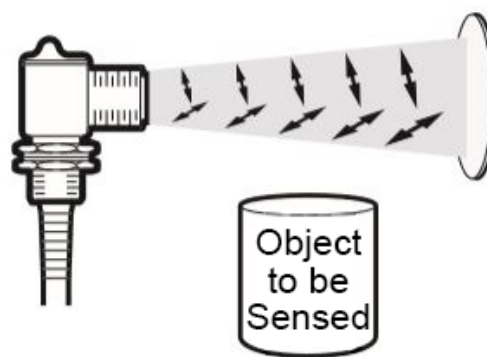
1. The sensors are typically held in place by a jam nut.



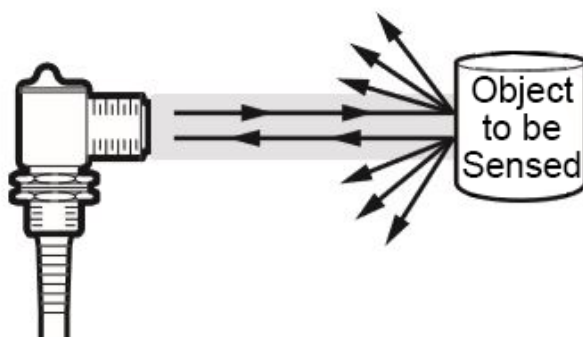
2. The sensor must not have any obstructions or debris in front of it.
3. Keep sensor heads, lenses, and reflectors clean.
4. The sensors must be adjusted according to the manufacturer specifications as sensing distances vary by make and model.
5. The following sensor types are typically deployed on Blueprint Automation equipment.



Transmitted Beam



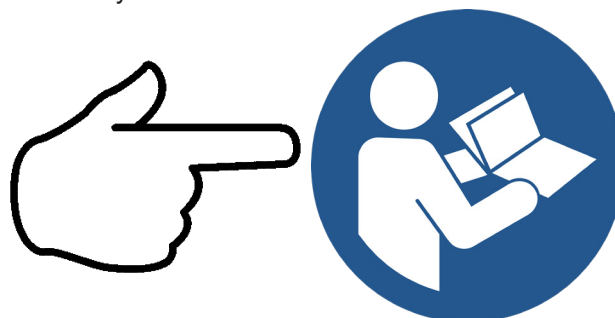
Polarized Retroreflective



Sharp Cutoff Diffuse

Sensor Replacement

When replacing defective sensors it is advisable to consult the instruction bulletins furnished with the new sensors by the OEMs.

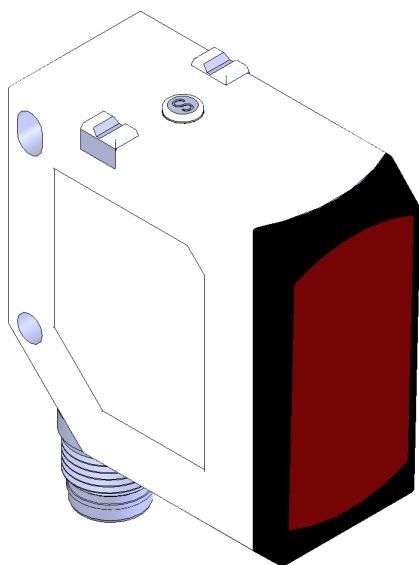


Clean The Laser Sensors

Build-up of dirt on the sensor's lenses may affect the efficacy of the devices. It is therefore recommended to perform maintenance regularly. The laser sensors are EN/IEC 60825-1 Class 1 laser products, that are considered safe under reasonably foreseeable conditions of operation, including the use of optical instruments for intra-beam viewing.



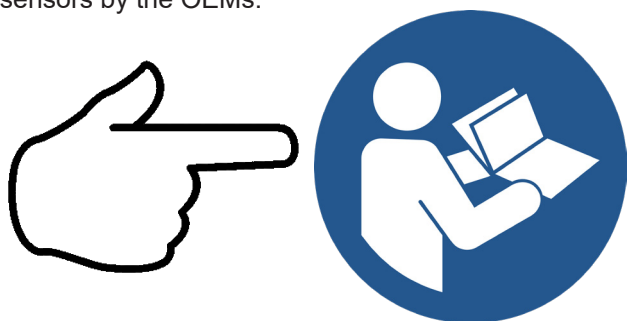
It is recommended to use compressed air to blow dust away. Clean with a dry or a damp cloth using a mild detergent if necessary. Do not use any abrasive or solvent cleaners.



Enclosure is rated IP67 according to IEC/EN 60529

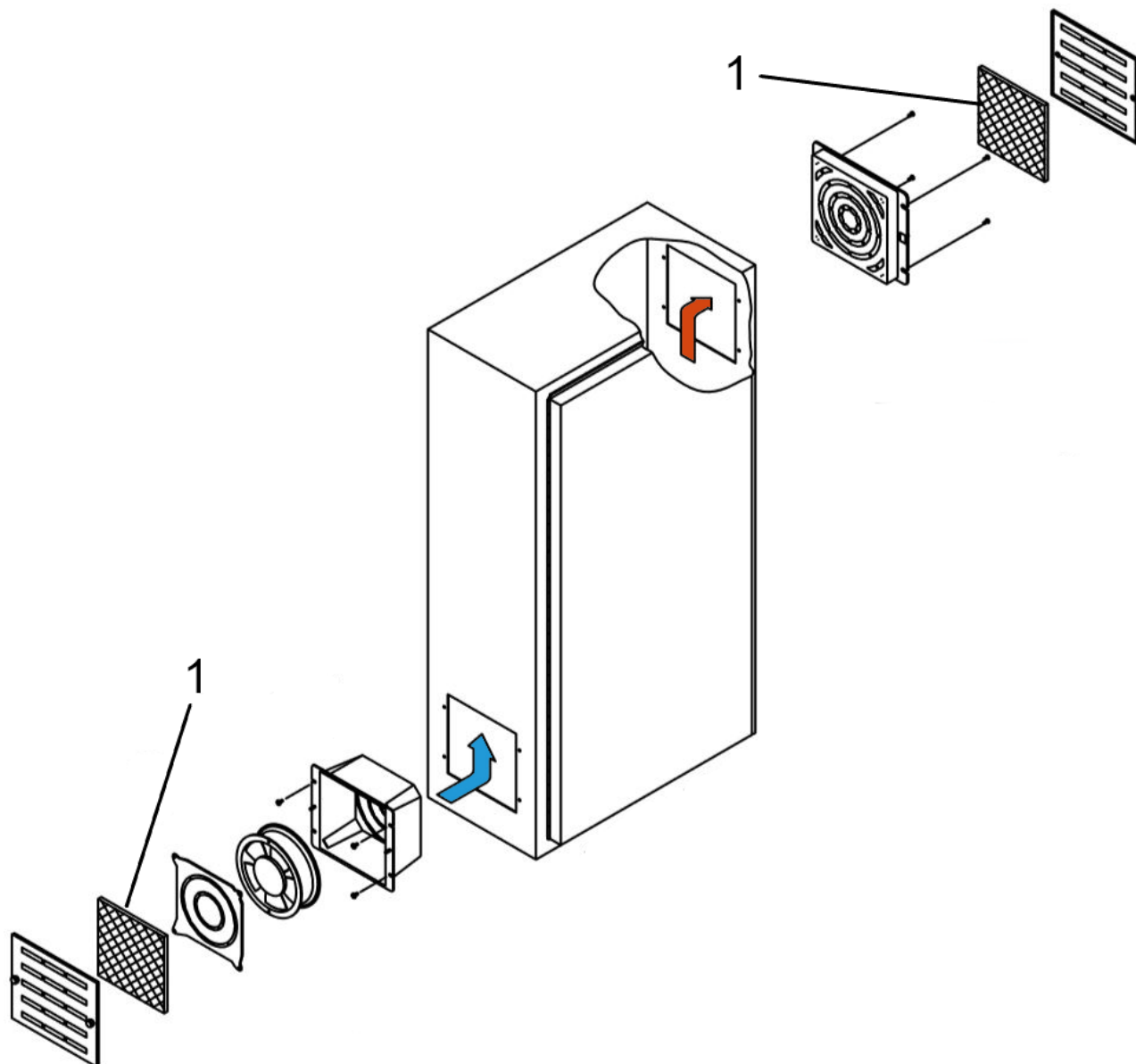
Sensor Replacement

When replacing defective sensors it is advisable to consult the instruction bulletins furnished with the new sensors by the OEMs.



Cabinet Filters Maintenance

In order to maintain optimal performance and prevent damage to temperature sensitive components, it is necessary to exchange heated air within the cabinet with relatively cooler air from without; this objective is achieved by means of a cooling fan and exhaust grille. Filters (1) are located at both the fan and exhaust ports to preclude airborne particulate matter from getting into the electrical and control cabinets. These filters (1) should be cleaned or changed (see note below) regularly to maintain requisite cooling air flow through the cabinet(s).



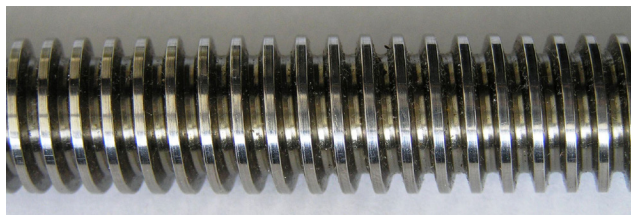
Simply snap open the hinged grilles for easy filter (1) access.

NOTICE

***The NEMA type 1 foam filter media is washable.
The NEMA type 12 non-woven "disposable" filter
is not washable and must be replaced.***

Lead-screws Maintenance

The correct lubrication of lead-screws is not only a prerequisite for the achievement of the theoretical lifetime, but also has an influence on smooth running properties, heat generation during operation and on the idling torque.

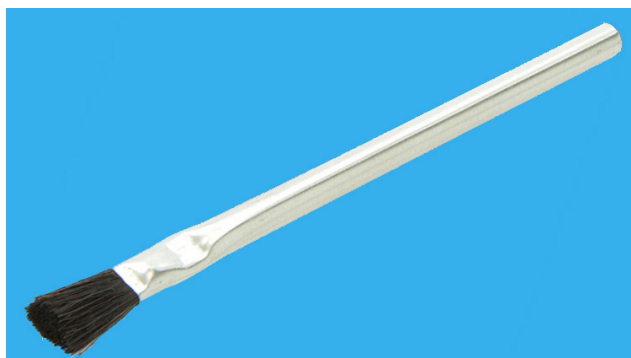


Lubrication

Grease lubrication is the usual lubrication for trapezoidal thread screws. The lubrication intervals depend on the operating conditions; it is recommended that the shaft is cleaned before being greased. Grease grades: food grade roller bearing greases **without a solid lubricant constituent**. Apply with a non-metallic brush.

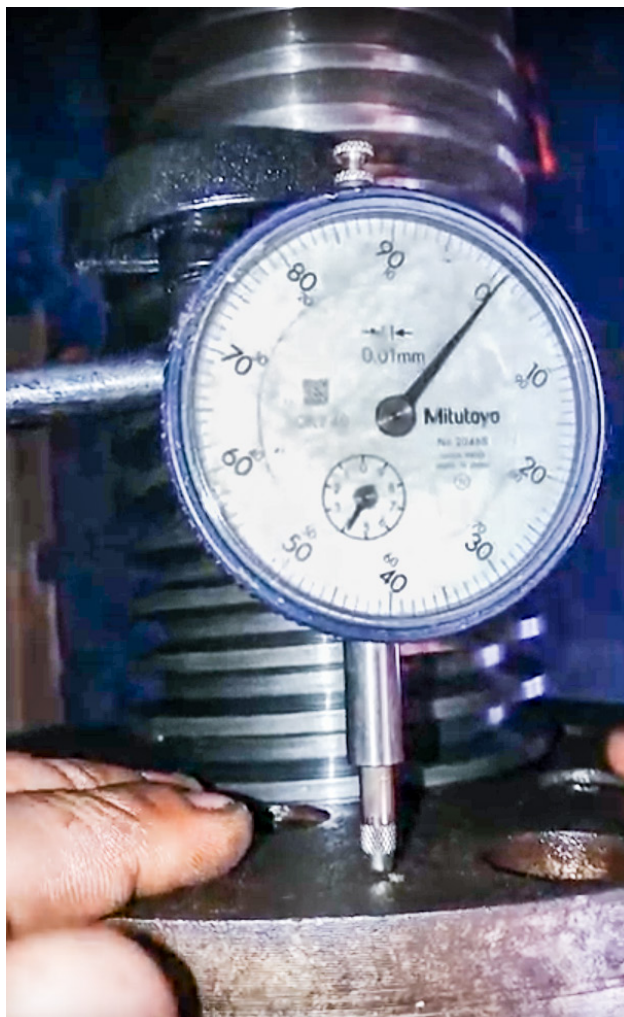
Lubrication specifications:

- NLGI 2 grease such as Henkel LOCTITE® LB 8104 is recommended.



Wear

The unit may be tested by hand; the nut must be replaced if the axial play of a single thread lead screw is greater than 1/4 of the lead.



Asynchronous Gear-motor Maintenance

Asynchronous (or induction) gear-motors should be inspected regularly for leaks, cracks, and external damage.

Check for anomalous running noises; gear-heads exhibiting unusual running noises and/or vibrations could indicate damage to the gear-head. In this case the gear-head should be shut down and a general overhaul carried out.

NOTICE

Important: The general overhaul must be carried out by qualified personnel in a specialised workshop with appropriate equipment in observance of applicable legal obligations.

Checking The Oil Level

Check the oil level with an oil temperature of between +20 °C...+40 °C. Extract the oil level hole plug, and check the oil level. The correct oil level is at the lower edge of the oil level hole. Top off with the correct oil type (Mobil synthetic gear oil type SHC 636 [ISO 6743 VG 680]) as necessary. Reinstall the oil level hole plug and tighten to the correct torque.

Changing The Oil

To change the oil in the gear-head, proceed as follows:

1. Place a catchment vessel under the oil drain screw.
2. Unscrew and extract the oil level hole plug.
3. Unscrew and extract the oil drain plug.
4. Drain all the oil from the gear-head.
5. Inspect the seal ring and plug, replace worn or damaged seal and plug if threads are damaged.
6. Reinstall the drain plug with seal into the drain hole and tighten to the correct torque.
7. Using a suitable filling device, refill with oil of the same type through the oil level hole (or through the pressure vent hole) until oil emerges through the oil level hole.
8. Wait at least 15 minutes and then check the oil level. Adjust oil level if necessary and reinstall the oil level hole plug to the correct torque.

The locations of the vent screw, drain plug, and oil level plug will vary by specific gear-head model and mounting orientation, reference the following graphics to find the specific arrangement for the unit under service.

WARNING

Residual burn hazard due to hot oil: Allow the gear-motor to cool down before carrying out maintenance or repair work. Wear protective gloves.

WARNING

Particles or liquids thrown up during servicing and maintenance can be injurious. Use caution when cleaning with compressed air or water. Use protective eye-wear.

WARNING

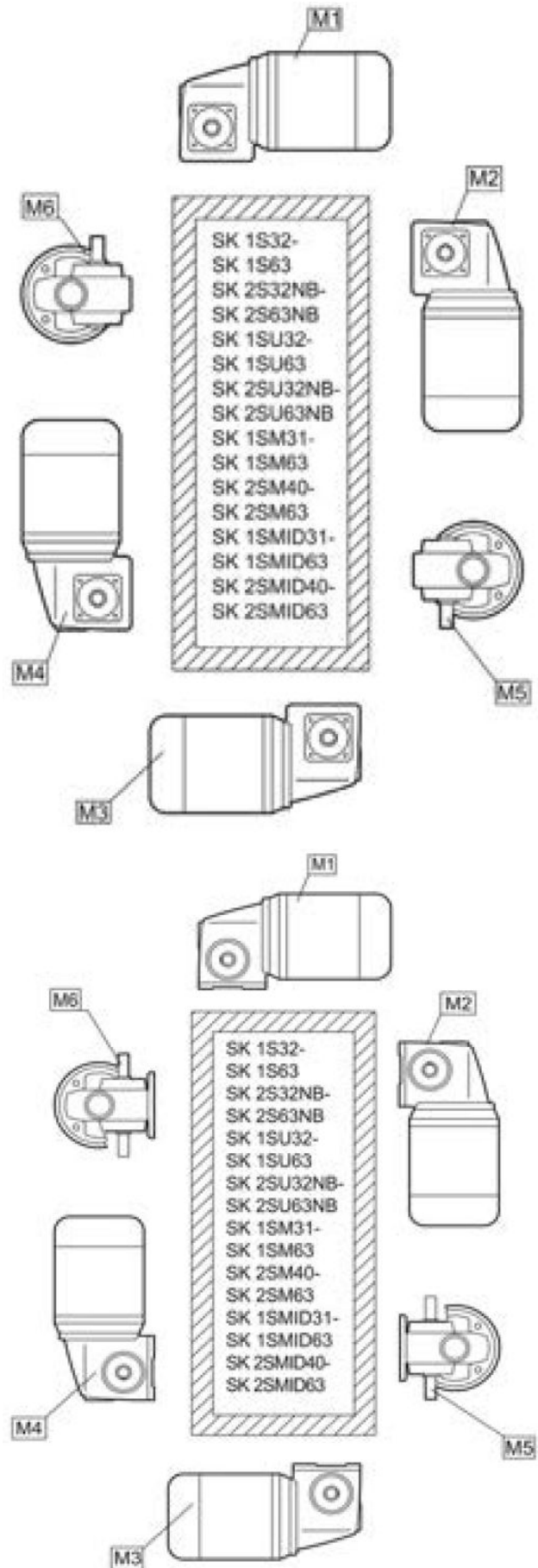
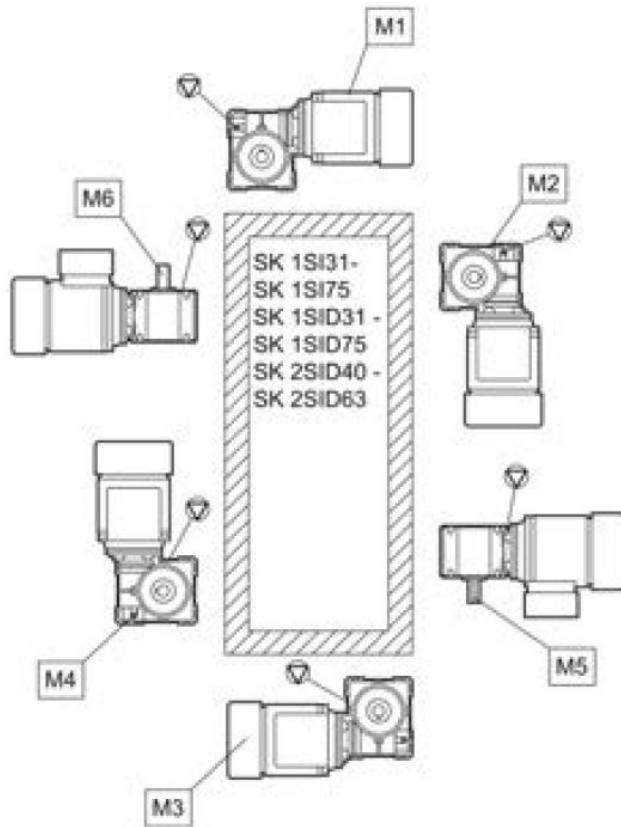
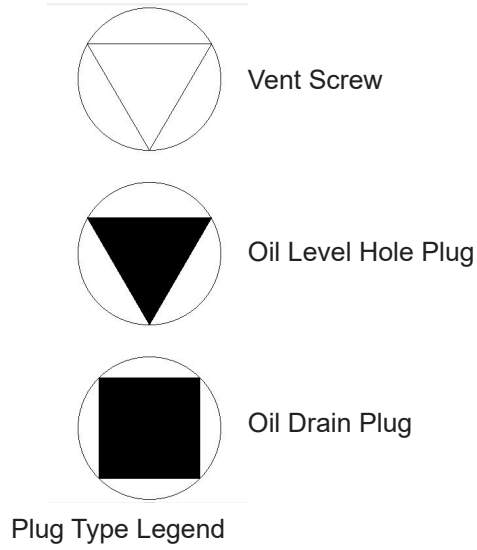
Whenever attempting to service or install this type of product, it is strongly recommended that the motor be put in a safe condition (power off) before servicing. Failure to follow the instructions in this manual may result in product damage, equipment damage, and serious or fatal injury to personnel.

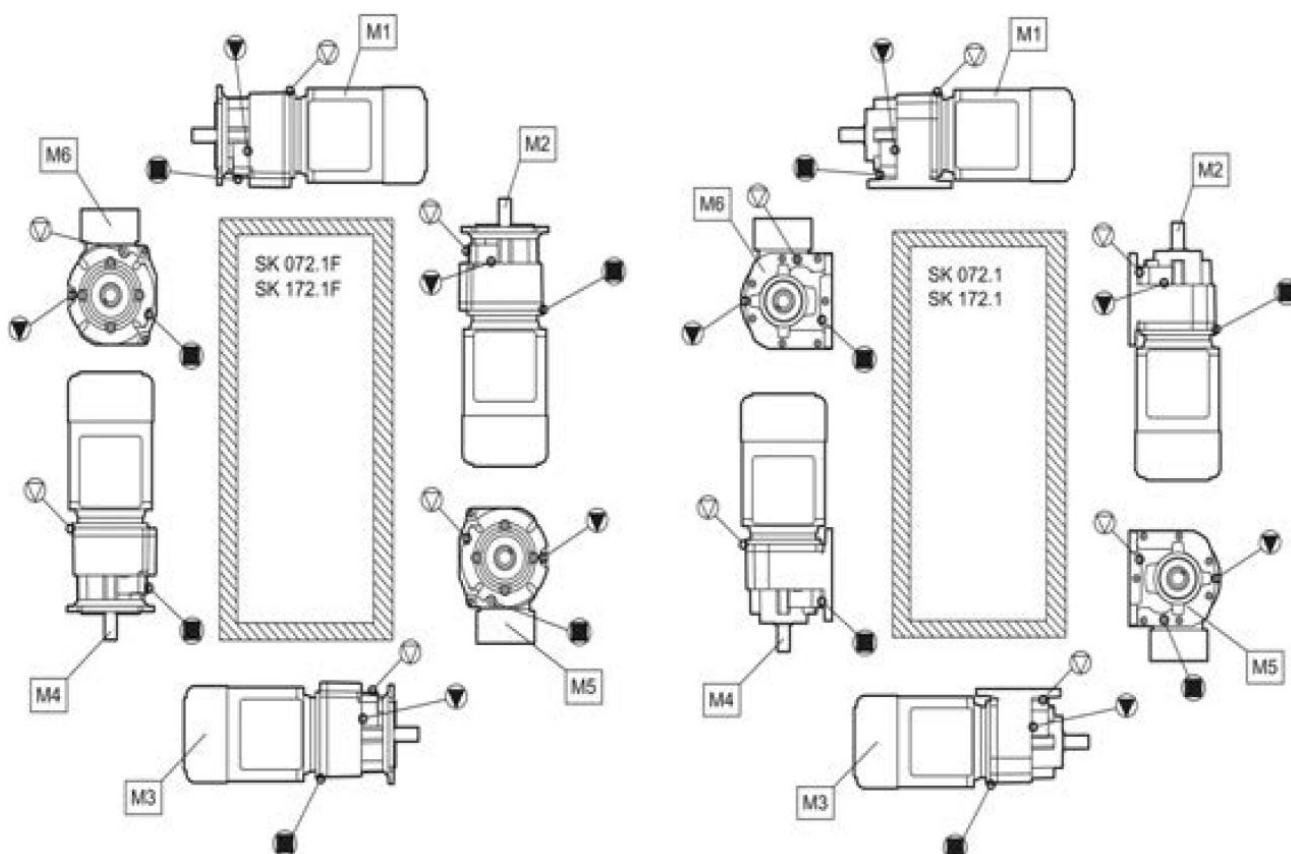
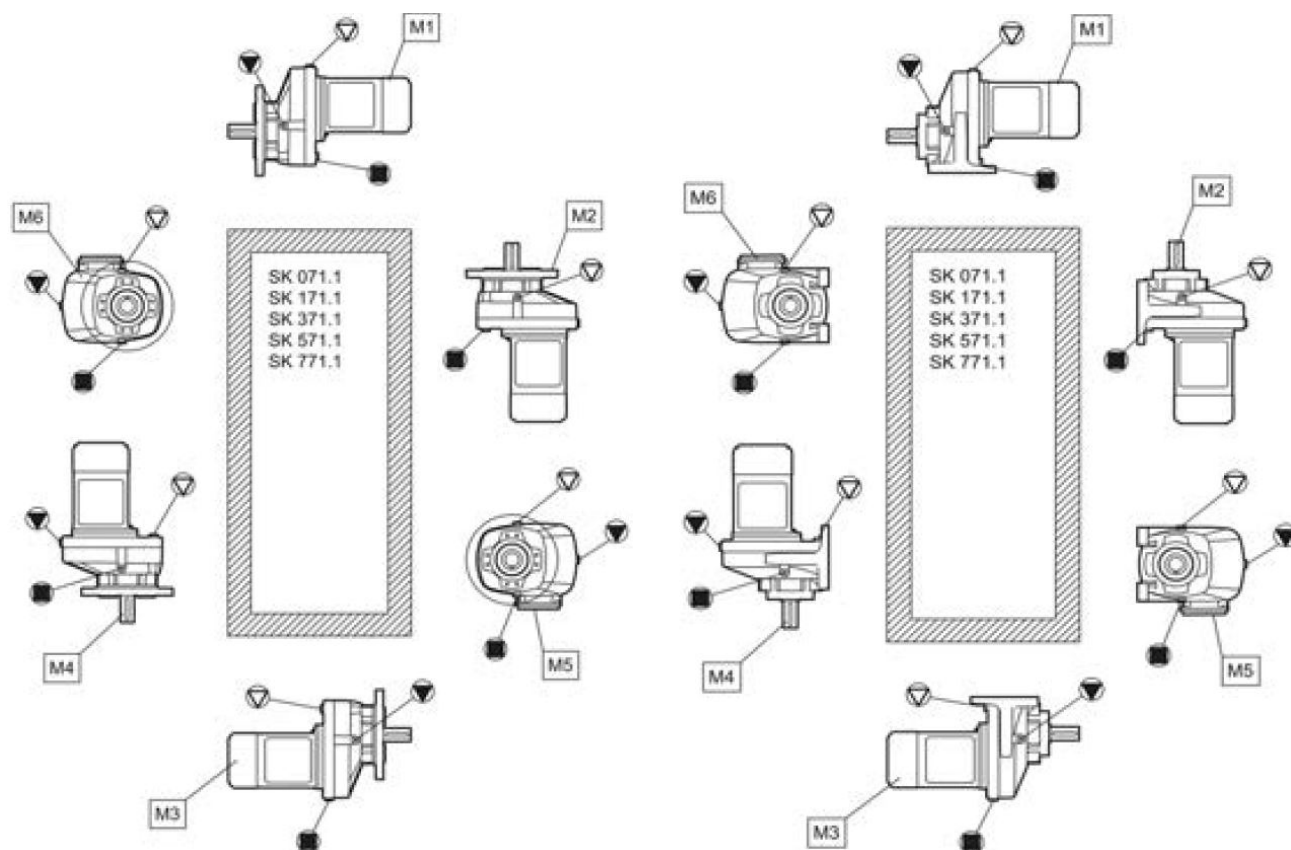
NOTICE

Important: Take care that no dirt or water enters the shaft sealing rings or the vents when cleaning. Dirt or water in the shaft sealing rings may cause leaks.

Bolt torque [N•m]						
Size	Property Class			Cover screws	Threaded pin on coupling	Screw connexions on protective covers
	8.8	10.9	12.9			
M4	3.2	5	6	-	-	-
M5	6.4	9	11	-	2	-
M6	11	16	19	-	-	6.4
M8	27	39	46	11	10	11
M10	53	78	91	11	17	27
M12	92	135	155	27	40	53
M16	230	335	390	35	-	92
M20	460	660	770	-	-	230
M24	790	1150	1300	80	-	460
M30	1600	2250	2650	170	-	-
M36	2780	3910	4710	-	-	1600
M42	4470	6290	7540	-	-	-
M48	6140	8640	16610	-	-	-
M52	9840	13850	24130	-	-	-
G1/2	-	-	-	75	-	-
G3/4	-	-	-	110	-	-
G1	-	-	-	190	-	-
1 1/4	-	-	-	240	-	-
G1 1/2	-	-	-	300	-	-

Oil Plug Type Locator Drawings



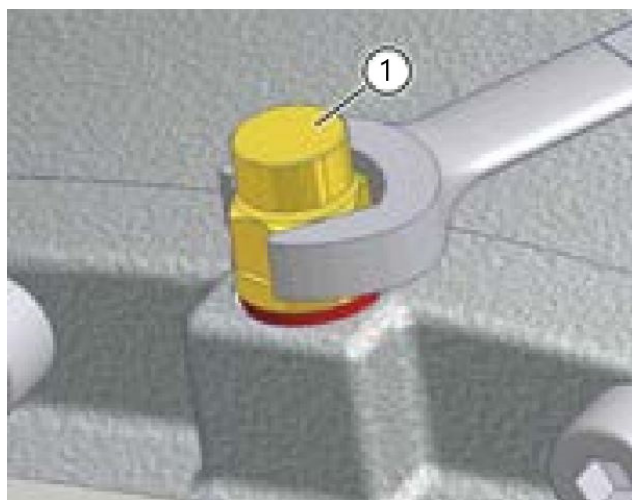


Re-lubricating the Bearings

For bearings which are not oil-lubricated, and whose holes are completely above the oil level, replace the roller bearing grease (recommended grease: Klüber PETAMO GHY 133 N).

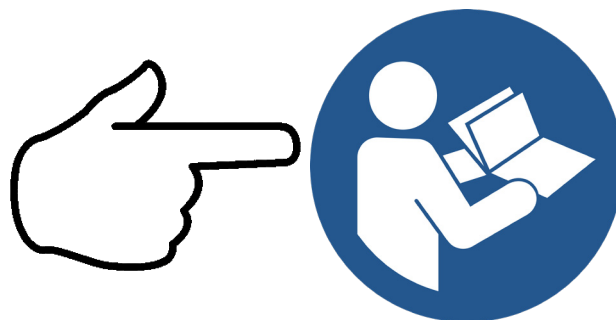
Cleaning or Replacing the Vent Screw

Unscrew the vent screw (1) and thoroughly clean (e.g., with compressed air) and fit the vent screw in the same place, if necessary use a new vent screw with a new seal ring.



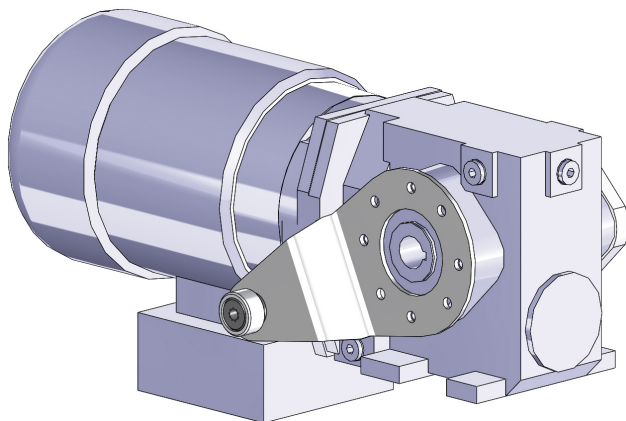
Gear-head Replacement

When replacing defective gear-heads it is advisable to consult the installation instruction bulletin furnished with the new Gear-head by the OEM.



Replacing the Shaft Sealing Ring

Once the shaft sealing ring has reached the end of its service life, the oil film in the region of the sealing lip increases and a measurable leakage with dripping occurs. The shaft sealing ring must then be replaced. The space between the sealing lip and the protective lip must be filled approximately 50% with grease on fitting. Take care that after fitting (recommended grease: Klüber PETAMO GHY 133 N), the new shaft sealing ring does not run in the old wear track.



Asynchronous Gear-motor